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PAEAR: Point Clouds Area Exploration and Active Recognition method driven by reinforcement learning for robotic welding

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ABSTRACT

For welded components without CAD models or prior geometry, robots face significant challenges in translating and interpreting design information into welding-relevant knowledge. When facing unfamiliar workpieces, welding robots cannot autonomously understand seam, edge, and surface elements. Consequently, welding seam localization and trajectory planning still rely heavily on manual teaching and experience-based search. In this study, a weld local feature descriptor (W-LFD) is developed to capture the local patterns of typical weld joints using interpretable geometric statistics, including inter-cluster surface angle, local density coefficient of variation, and normalized dispersion ratio. Together with small-sample pre-training, a prior feature library is established by coupling the descriptor with its parameters. Moreover, weld area acquisition is formulated as an active exploration problem in point clouds space. A reinforcement learning-based point clouds exploration and active recognition (PAEAR) method is proposed with a translation–scaling multi-level action strategy. This design enables identification and segmentation of regional point clouds for different welding seam categories in part-level scenes. Experimental results show that, across 2 simulated and 2 real-world scenarios, PAEAR achieves an mIoU of 86.86% using only 4 labeled samples, improving upon mainstream semantic segmentation methods by up to 81.22%, while reducing the area recognition error rate to 5.4%. The proposed method can serve as a front-end perception module for welding systems, providing reliable regional point clouds inputs for downstream welding seam localization and welding trajectory planning, while reducing dependence on manual teaching and prior models.

1. Introduction

The advent of smart manufacturing has driven a transformation in manufacturing processes, with low-code and no-code programming significantly enhancing the agility of production line reconfiguration [1]. Consequently, the field of robotics is experiencing a profound paradigm shift from traditional automation to true autonomy, propelled by advancements in embodied intelligence [2–6]. This evolution marks a seminal transition from a paradigm of being told physical laws to one of independently learning them [7], aiming to equip robots with human-like autonomous capabilities in unstructured environments [8].

In order to achieve such autonomy, manufacturing systems fundamentally require robust perception capabilities to identify, understand, and adapt to unknown or uncertain workpieces without relying on prior knowledge. This vital requirement for autonomous perception naturally exposes significant challenges in specific applications such as robotic welding.

While existing automated welding systems perform well in large-scale standardized production, they still largely rely on pre-defined models or manual teaching. As a result, in complex high-mix low-volume scenarios where prior models are absent or workpiece conditions are highly uncertain, these conventional methods face severe limitations during the critical pre-welding stage. From an industrial perspective, CAD models are often unreliable or unavailable. In heavy

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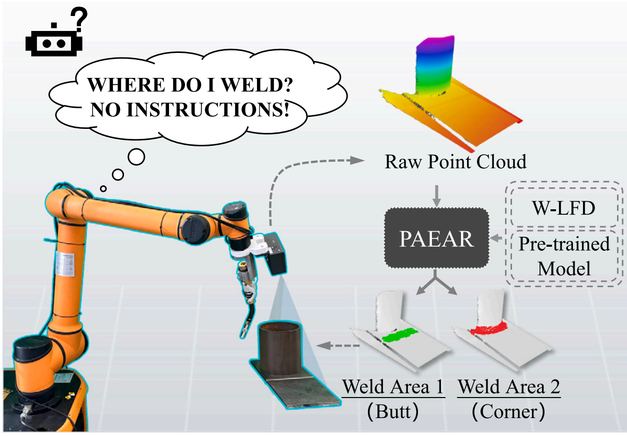


Fig. 1. Challenges faced by welding robots in interpreting unfamiliar parts and schematic diagram of autonomous weld recognition and classification based on PAEAR.

manufacturing, accumulated machining tolerances and thermal deformations render nominal CAD designs inaccurate [9]. In remanufacturing, original models of worn components are often unavailable [10]. Forced to rely on manual teaching for these unmodeled workpieces, programming time becomes a critical bottleneck. Studies indicate that in high-mix low-volume production, manual robot programming can account for over 70% of the total manufacturing cycle, making traditional automation economically unviable without robust autonomous perception [11]. As illustrated in Fig. 1, robotic welding system frequently lacks a practical front-end perception approach capable of rapidly transforming on-site acquired 3D point clouds into semantic information, including candidate welding seam areas and types, directly usable for localization and path planning [12,13]. Autonomous machines, unlike skilled human operators, lack the capability to interpret features of unfamiliar parts and to formulate strategies based solely on visual observation [14,15].

Significant advancements have been made in robotic weld positioning algorithms, with sensor-based feature extraction and real-time weld localization being widely adopted in the industry [16]. Concurrently, welding programming methods integrated with sensor technologies have achieved notable success in addressing dimensional deviations and seam searching [17]. However, despite the maturation of positioning algorithms for single-type welds [18,19], substantial gaps persist in the active perception and decision-making phase preceding precise localization. This current semi-autonomous paradigm indicates a lack of macro-level understanding of the welding object [20]. Consequently, when simultaneously dealing with multiple discrete weld joint types, integrating efficacious positioning algorithms to accurately localize all weld types remains a formidable challenge [21,22].

Current welding seam recognition methods for unmodeled workpieces primarily rely on geometric analysis of point clouds and can generally be categorized into two strategies: differential-geometry-based feature extraction and clustering-based segmentation. The former focuses on locating welding seam centers using local geometric attributes. For example, Zhang et al. [23] and Liu et al. [24] employed the LOBB operator and an edge-intensity response method, respectively, to efficiently extract fold points and salient feature points with accuracy better than 1 mm. For more complex multi-pipeline intersection models, Liu et al. [25] introduced discrete curvature features to achieve a fitting accuracy on the order of 0.1 mm. In contrast, clustering-based strategies emphasize holistic segmentation of weld areas through data-driven mining. DBSCAN-based clustering has shown good adaptability to various welding joint types, with detection errors as low as 0.11 mm [26]. Similarly, offline extraction pipelines based on point clouds segmentation are capable of limiting the maximum error to within 1.9 mm [27]. Although these methods can effectively identify boundary lines and junction lines in complex point clouds and can address localization in some scenarios, they often oversimplify contextual characteristics of welding areas by describing only geometric edge cues. As a result, it is difficult to simultaneously achieve both complete detection and accurate localization. Moreover, rule-based segmentation is prone to boundary ambiguity in areas with intersecting features, and neighborhood-based analysis on complex point clouds incurs high computational cost, leading to low efficiency.

Data-driven approaches have shown great potential for feature segmentation tasks, and the introduction of deep learning has provided new perspectives for point clouds processing and feature recognition in welding applications [28,29]. Deep-learning-based welding seam detection methods generally exhibit strong capability in learning complex features. Early supervised learning studies primarily focused on optimizing network architectures for welding seam recognition and segmentation. For example, an improved Faster R-CNN model with enhanced interpretability [30] and a dedicated segmentation network [31] were proposed for point clouds recognition and segmentation. However, in high-mix low-volume welding, acquiring large-scale annotated point clouds across multiple weld joint types is costly, rendering such models vulnerable to limited labeled data; consequently, research has increasingly shifted toward self-supervised learning. For example, a self-supervised machining feature recognition framework that can be trained on unlabeled data was proposed [32]. Meanwhile, reinforcement learning-based point clouds exploration methods have gradually been proposed for target recognition and planning [33,34]. In recent years, physics-data fusion has emerged as a promising direction in manufacturing research: mechanistic priors can improve interpretability and, at the same time, enhance generalization under small-sample conditions. Therefore, for localization in unknown workpieces with multiple welding joint types, there is an urgent need for an active recognition approach that integrates interpretable geometric priors with data-driven learning.

In order to address the challenge of actively discovering and interpreting multi-type welding seam areas in point clouds of unknown workpieces, this paper proposes a reinforcement learning-based method. The method for multi-type welding seam area recognition is guided by geometric priors, especially in scenarios where annotated samples are scarce. The main contributions are summarized as follows:

(1) An interpretable weld local feature descriptor (W-LFD) is designed to parameterize the local geometric and statistical patterns of point clouds for typical joint types, and a reusable prior feature library is constructed using a small number of samples.
(2) The acquisition of weld area is reformulated as an area exploration problem in point clouds space, with a multi-level action exploration strategy. Geometric priors are incorporated into policy learning via state and reward design, along with motion guidance. This enhancement leads to improved coverage efficiency in unfamiliar welding scenarios and enhances recognition and classification robustness under small-sample conditions. The framework provides front-end perceptual inputs for downstream localization and trajectory planning.

The remainder of this paper is organized as follows: Section 2 introduces the construction of the weld local feature descriptor; Section 3 presents the reinforcement learning formulation and recognition pipeline; Section 4 reports the experimental setup and result analysis; and Section 6 concludes the paper and discusses future work. The overall workflow of the proposed algorithm is illustrated in Fig. 2, where Flow 1 denotes the pre-training process and Flow 2 denotes the active recognition process.

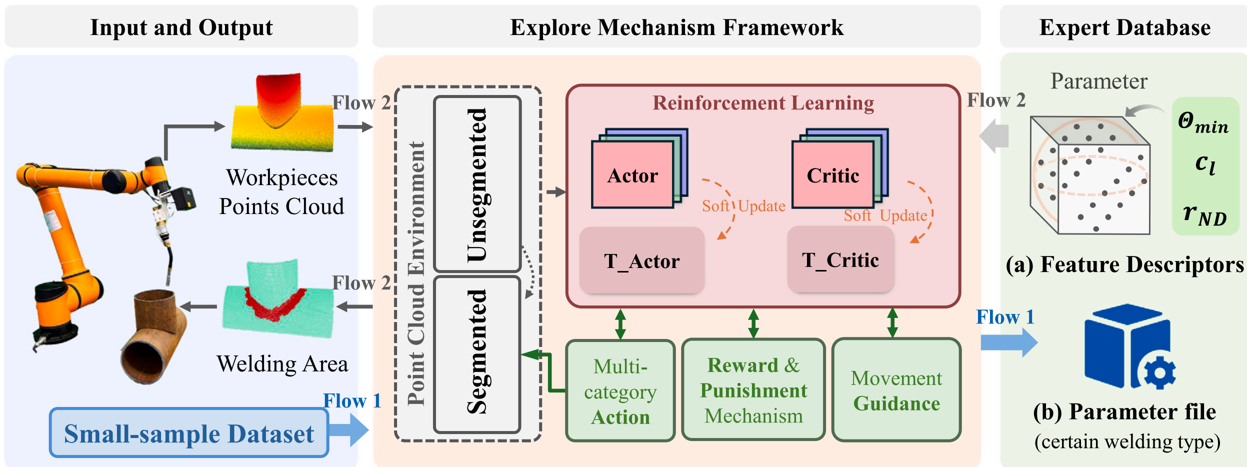


Fig. 2. Overall workflow of the proposed algorithm. Flow 1 denotes the generation of parameter files from small-sample datasets, while Flow 2 illustrates the perception process mapping input point clouds to segmented welding areas.

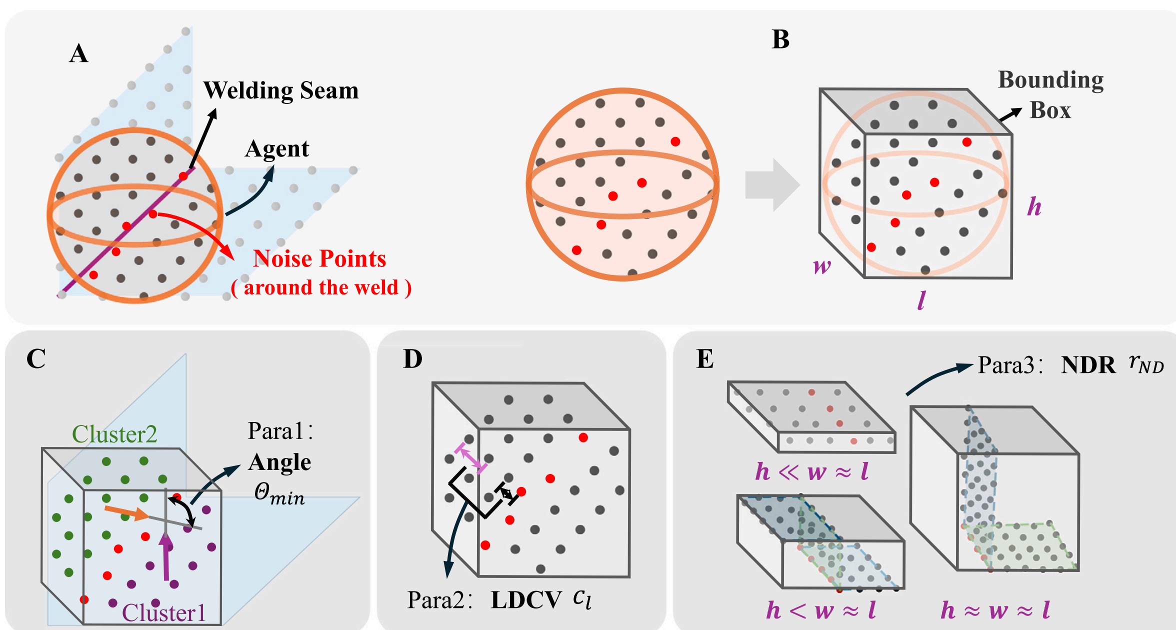


Fig. 3. Weld local feature descriptor (W-LFD).

2. Weld local feature descriptor

A point clouds bounding box refers to a geometric box that completely encloses three-dimensional point clouds data. It enables efficient processing and analysis of point clouds data through simplified geometric representation [35]. As shown in Fig. 3A and B, an oriented bounding box (OBB) is adopted to construct a local point clouds feature descriptor for welding seam characterization, termed the weld local feature descriptor (W-LFD). This descriptor is used to represent the geometric properties and mathematical characteristics of discrete welding seam point clouds and can accurately capture multiple aspects of welding seam information.

Weld local feature descriptor (W-LFD) is a transformation tool that maps point clouds data into welding feature information by parameterizing the local geometric and statistical patterns of typical welded joints. In this manner, unstructured spatial coordinates are converted into explicit features with clear physical interpretability. In order to accurately characterize the local features of welding seam point clouds, three parameters are defined in this study: the inter-cluster surface angle, the local density coefficient of variation (LDCV), and the normalized dispersion ratio (NDR) [36]. Fig. 3 illustrates the modeling schematic of W-LFD. Intuitively, the inter-cluster surface angle captures the relative orientation between adjacent surface patches and reflects the geometric joint configuration, the LDCV quantifies local point density fluctuations caused by gaps or missing material at weld interfaces, and the NDR describes the overall shape anisotropy of the local bounding box to distinguish different weld joint morphologies.

2.1. Inter-cluster surface angle

As demonstrated in Fig. 3C, the point clouds contained within the specified bounding box has been processed using the K-means algorithm to create a cluster. Different weld joint types are determined by the curvature of the region represented by the inter-cluster angle. This approach is capable of representing welding seam information for a variety of joint angles, including butt, lap, and corner joints. The set of points enclosed by the agent is denoted as $S(a, r)$, where a is the agent center vector and r is the agent radius; each point is denoted as

p, with $\forall, p \in S(a,r)$. To describe the inter-cluster surface angle, the local normal $n(p)$ of points within the agent enclosure is first computed using Eq. (1), followed by the neighborhood covariance matrix C_p. Finally, the computed minimum inter-cluster angle $\Theta_{\min}$ is taken as the inter-cluster surface angle.

$$n(p) = \arg\min_{\|n\|=1} n^{\top} C_p n \tag{1}$$

where n denotes a unit vector or a candidate direction vector and is treated as the optimization variable, and the computation of C_p is given in Eq. (2).

$$C_p = \frac{1}{|N_p|} \sum_{y\in N_p} (y - \frac{1}{|N_p|} \sum_{y\in N_p} y)(y - \frac{1}{|N_p|} \sum_{y\in N_p} y)^{\top} \tag{2}$$

where N_p denotes the radius-based neighborhood of point p, and y denotes a neighboring point within this neighborhood.

The minimum inter-cluster surface angle $\Theta_{\min}$ is calculated using Eq. (3).

$$\Theta_{\min}(a,r) = \min_{1\le k<\ell\le m} \arccos\left(\mathrm{clip}(u_k^{\top} u_l, -1, 1)\right) \tag{3}$$

$$u_k = \frac{\sum_{p\in C_k} n(p)}{\left\|\sum_{p\in C_k} n(p)\right\|} \tag{4}$$

The cluster-center normal of the k-th-th cluster, denoted as u_k, is defined in Eq. (4), where $k = 1, \ldots, m$ and m is the total number of clusters; similarly, u_l denotes the cluster-center normal of the lth cluster.

2.2. Local density coefficient of variation

This study observes that point clouds uniformity changes at welding seam interfaces due to the presence of gaps. Such changes are primarily manifested as point clouds disorder or partial point loss. To address this issue, a statistical parameter model termed the local density coefficient of variation (LDCV), denoted as c_l, is constructed to quantitatively characterize the gap features at the interface between the point clouds of two workpiece surfaces. The LDCV is capable of capturing local fluctuations in point clouds density. Compared with uniformly distributed surface regions, the point clouds at gap interfaces exhibits significantly higher LDCV values due to missing or disordered points. The parameter c_l is dimensionless and is used to measure the degree of dispersion in local point clouds density. As shown in Eq. (5), it is mathematically defined as the ratio of the standard deviation to the mean of the local point density.

$$c_l = \frac{\sigma_d}{\mu_d} = \frac{\sqrt{\frac{1}{N_{in}} \sum_{p_i\in R} (d_i - \mu_d)^2}}{\frac{1}{N_{in}} \sum_{p_i\in R} d_i} \tag{5}$$

where c_l is calculated as the ratio of the standard deviation of the local point density within the junction region (σ_d) to the mean local point density (μ_d). Here, d_i denotes the local density of point p_i, and N_{in} represents the total number of points currently encompassed within the region. A higher c_l value indicates a more pronounced gap characteristic. The LDCV is highly sensitive to sparsity and disorder within the point clouds, enabling it to effectively capture density variations induced by the presence of gaps; consequently, the metric demonstrates strong robustness.

The LDCV serves as a critical bridge between unstructured visual data and actionable welding semantics. In real-world manufacturing, the physical interface between two assembled workpieces inevitably contains assembly tolerances, grooves, or joint clearances. Rather than treating these geometric imperfections as sensor noise, the LDCV explicitly utilizes them as a physical signature for welding seam localization. This ensures that the robot's exploration policy is driven by actual joint morphology and physical assembly characteristics, rather than being misled by unmodeled background anomalies.

2.3. Normalized dispersion ratio

When the agent covers local welding seam point clouds corresponding to different welding types, the resulting bounding boxes exhibit distinct characteristics. These variations are reflected in the shapes of the bounding boxes. For butt joints, the bounding box takes the form of a flat cuboid with a large length-width ratio and a relatively small height. In the case of corner joints, the bounding box approximates a cuboid with three edges of comparable lengths. By contrast, for lap joints, the bounding box is characterized by a relatively small height compared to its length and width. Accordingly, the edge lengths of the bounding box are defined as w, l, and h. Based on the statistical properties of these edge lengths and their proportional relationships, a normalized dispersion ratio (NDR), denoted as r_{ND}, is constructed to distinguish different bounding box shapes. The computation of r_{ND} is given in Eqs. (6) and (7).

$$\sigma_{sl} = \sqrt{\frac{(x_1-\mu_{sl})^2 + (x_2-\mu_{sl})^2 + (x_3-\mu_{sl})^2}{3}} \tag{6}$$

$$r_{\mathrm{ND}} = \frac{\sigma_{sl}\cdot x_3}{\mu_{sl}\cdot x_2} \tag{7}$$

$$\mu_{sl} = \frac{x_1+x_2+x_3}{3} \tag{8}$$

Here, σ_{sl} denotes the standard deviation of the side lengths, and μ_{sl} represents the mean side length, which is calculated according to Eq. (8). For this calculation, the dimensions w, l, and h are sorted and re-designated such that $x_1 \le x_2 \le x_3$. Based on this metric, the study establishes the following classification criteria: when the bounding box yields $r_{\mathrm{ND}} < 0.1$, the side lengths exhibit high uniformity, indicating the enclosed point clouds resembles a corner joint. When $0.1 \le r_{\mathrm{ND}} \le 0.7$, the maximum dimension is proximate to the median dimension, suggesting the point clouds corresponds to a lap joint. Finally, when $r_{\mathrm{ND}} \ge 0.7$, the maximum dimension is significantly larger than the median dimension, identifying the point clouds as characteristic of a butt joint.

3. Active recognition method

3.1. Point clouds exploration mechanism

This paper proposes a point clouds exploration mechanism based on a reinforcement learning framework. The framework is predicated on the conventional actor-critic network architecture as its fundamental algorithmic foundation. The subsequent construction of a decision-making system drives an agent to perform autonomous exploration. The system operates within a three-dimensional point clouds environment. To better accommodate the characteristics of point clouds data processing and the requirements of efficient exploration, the proposed method abandons the traditional paradigm in which an agent relies solely on a single state representation. Instead, the attributes of three-dimensional point clouds bounding boxes are incorporated into the state perception and decision-making process of the agent. This establishes a novel agent endowed with awareness of point clouds bounding box properties. Furthermore, to address the issues of disordered exploration trajectories and incomplete spatial coverage in point clouds space exploration, a multi-level action strategy is introduced to significantly refine and optimize the motion decision space of the agent. As illustrated in Fig. 4, the overall framework integrates the bounding-box-aware agent with the multi-level action strategy.

During the pre-training stage, the network was trained on a hybrid point clouds dataset consisting of simulated 3D camera scans of CAD models in a virtual environment and real 3D camera scans of physical workpieces. This hybrid training strategy endowed the model with strong generalization capability. The resulting trained model parameters were then used as the output of the pre-training process.

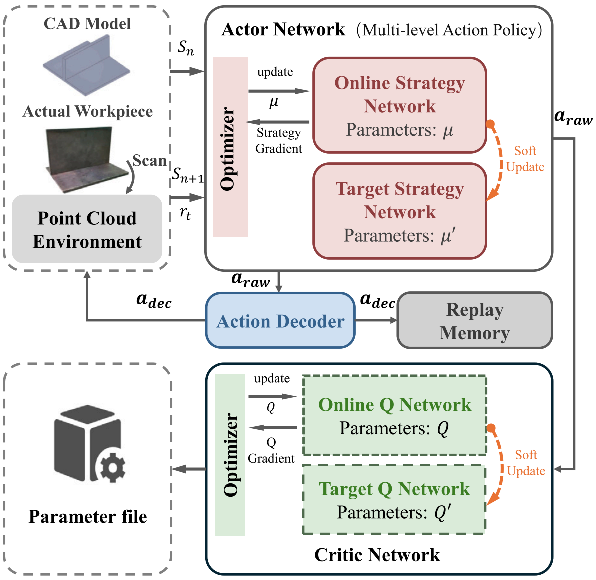


Fig. 4. Pre-trained network architecture diagram.

However, in the actual active recognition stage, the system is fully decoupled from the CAD models, and the only input is the point clouds data acquired by an on-site 3D camera from previously unmodeled workpieces.

To endow the agent with comprehensive perceptual capability of the welding environment, the state at time step t, denoted as s_t, is defined in Eq. (9) as a high-dimensional feature vector.

$$s_t = [p_t, r_t, \mathcal{F}_{\text{geo}}, h_{\text{visit}}] \tag{9}$$

$$\mathcal{F}_{\text{geo}} = \left[\Theta_{\min}, c_l, r_{\text{ND}}\right] \tag{10}$$

where $p_t \in \mathbb{R}^3$ and $r_t \in \mathbb{R}$ denote the normalized center coordinates and the perception radius of the agent, respectively. The local geometric feature vector $\mathcal{F}_{\text{geo}}$, defined in Eq. (10), is computed in real time based on the W-LFD method introduced in Section 2 and is used for semantic discrimination. In addition, the variable h_{visit} represents the historical visitation density of the current region, which is incorporated to prevent redundant exploration.

The point clouds exploration mechanism constructs a multi-level action-based motion strategy through an action decoder. By means of a modality selection mechanism, the action decoder ensures that the agent can jointly balance global exploration and local fine-grained analysis. During the learning process, the agent dynamically adjusts its policy according to state feedback. In the initial stage, when multiple reward signals remain at relatively low values, the agent tends to maintain the translation mode, performing broad traversal over the entire point clouds via positional transformations under a large scaling factor. Once potential welding seam features are detected, the agent automatically switches to the scaling mode to enable fine-grained focusing. Subsequently, upon the convergence of the scaling factor to the minimum value as determined by the workpiece's global bounding volume, the agent undertakes repetitive verification of the target area. The algorithm is terminated once the number of validations that satisfy the specific parameters of the weld local feature descriptor reaches a predefined threshold of 10. This process guarantees the accurate segmentation of the target point clouds.

In the proposed point clouds exploration mechanism, the raw action vector output by the actor is defined in Eq. (11), where η_t denotes the modality switching parameter and v_t represents the control parameter. The action decoder $\mathcal{D}$ maps the raw action into a physical environment action a_t^{dec} by means of an indicator function $\mathcal{I}(\cdot)$, as shown in Eq. (12). In this context, the symbol $\mathcal{T}(\cdot)$ represents the translation transformation, $\mathcal{S}(\cdot)$ denotes the scaling transformation, and the symbol δ is known as the modality switching threshold.

$$a_t^{\text{raw}} = [\eta_t, v_t]^T \tag{11}$$

$$a_t^{\text{dec}} = \mathcal{D}(a_t^{\text{raw}}) = \mathcal{I}(\eta_t < \delta) \cdot \mathcal{T}(\boldsymbol{v}_t) + \mathcal{I}(\eta_t \geq \delta) \cdot \mathcal{S}(\boldsymbol{v}_t) \tag{12}$$

The optimization objective of the critic network is to minimize the temporal-difference (TD) error. A key aspect lies in the computation of the target value, where the action output by the target actor μ' must first be transformed by the decoder $\mathcal{D}$, as shown in Eqs. (13) and (14).

$$L(\theta^Q) = \mathbb{E}_{(s_t, a_t^{\text{dec}}, R_t, s_{t+1}) \sim M}\left[\left(\hat{Q}_t - Q(s_t, a_t^{\text{dec}} | \theta^Q)\right)^2\right] \tag{13}$$

$$\hat{Q}_t = R_t + \gamma Q'(s_{t+1}, \mathcal{D}(\mu'(s_{t+1}|\theta^{\mu'}))|\theta^{Q'}) \tag{14}$$

The optimization objective of the actor is to adjust the parameters θ^μ such that the generated raw actions, after being decoded, can maximize the corresponding Q-value. The gradient update of the actor is expressed in Eq. (15).

$$\nabla_\theta \mu J \approx \frac{1}{N} \sum_{i=1}^{N} \nabla_a Q(s_i, a|\theta^Q)|_{a=a_i^{\text{dec}}} \cdot \nabla_{\theta^\mu} \mu(s_i|\theta^\mu) \tag{15}$$

where $L(\theta^Q)$ denotes the loss function of the critic network, $\mathbb{E}[\cdot]$ represents the expectation operator, and M denotes the replay memory. R_t represents the reward at time step t, and γ is the discount factor. The symbols Q and Q' represent the current critic network and the target critic network, respectively. Meanwhile, μ and μ' represent the current actor network and the target actor network. The parameters θ^Q, $\theta^{Q'}$, θ^μ, and $\theta^{\mu'}$ correspond to their respective networks, and $\nabla_{\theta^\mu} J$ denotes the policy gradient of the actor.

In the context of welding seam point clouds, a spherical agent construction method is proposed. The spherical agent is capable of analyzing the point clouds enclosed within its volume. After bounding box extraction and normalization, the enclosed point clouds is conveniently organized into a feature descriptor form for subsequent computation and analysis. Within the reinforcement learning framework, the design of an area recognition reward function is a critical step for guiding the spherical agent to learn effectively and optimize its policy. Eq. (16) presents the composition of the reward function, where R denotes the total reward of the algorithm and ω represents the weighting coefficients of the individual reward components. In this study, ω_1 and ω_2 are set to 0.4, and ω_3 is set to 0.2. The specific values of these weighting coefficients were determined through extensive empirical testing to balance accuracy and efficiency. The assignment of higher weights to coverage and exploration compared to feature matching is designed to ensure algorithmic efficiency; this configuration accelerates the overall exploration speed, encourages broader spatial traversal, and mitigates repeated local oscillations. Furthermore, since a rigorous multiple-verification mechanism is incorporated during the feature-matching stage, the assigned lower weight for feature matching is fully sufficient to provide effective guidance for semantic alignment. To achieve precise identification and comprehensive coverage of welding areas, a set of diversified reward function components is designed by taking the point clouds enclosed by the spherical agent as the quantitative basis. These components jointly account for both the extent and quality of the covered regions, ensuring that the agent can explore a larger portion of the point clouds while maintaining stable and focused coverage throughout the exploration process.

$$R = \omega_1 R_{\text{points}} + \omega_2 R_{\text{explore}} + \omega_3 R_{\text{match}} \tag{16}$$

In order to encourage the agent to cover a broader area, a convex hull is computed for the point clouds p within the radius r_{agent} of the agent. The convex hull is assumed to consist of a set of triangular faces, and each face is defined by its vertices. The convex hull volume

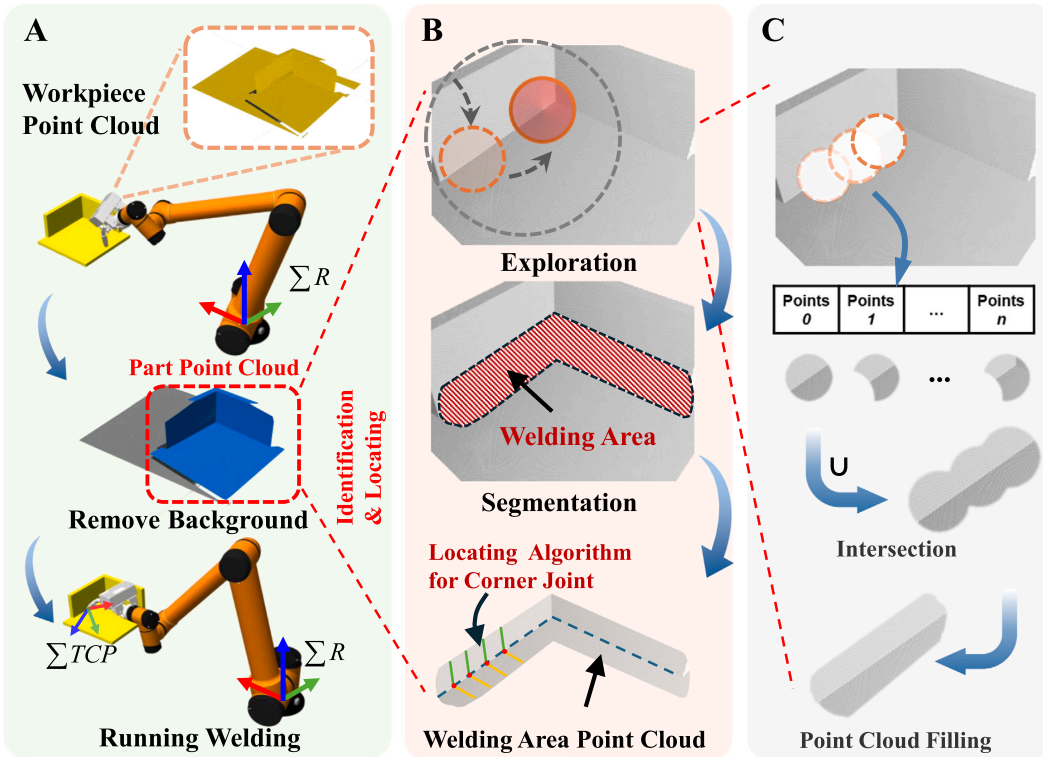


Fig. 5. Workflow of the proposed algorithm: from environment perception and active agent exploration to multi-step segmentation of welding seam point clouds. (A) Point clouds preprocessing and reinforcement learning environment construction. (B) Dynamic localization and identification of welding seam regions via the agent-based active exploration strategy. (C) Multi-step welding seam point clouds segmentation and region reconstruction.

is decomposed into a sum of tetrahedral volumes using a recursive or decomposition-based algorithm. The volume of each tetrahedron, denoted as V_{T_i}, is first computed using Eq. (17). The volumes of all tetrahedra are then summed in Eq. (18) to obtain the convex hull volume V_{hull}, which is subsequently normalized by the maximum convex hull volume $V_{\max}$.

$$V_{T_i} = \frac{1}{6}|\boldsymbol{v}_{i1} \cdot (\boldsymbol{v}_{i2} \times \boldsymbol{v}_{i3})| \tag{17}$$

$$V_{\mathrm{norm}} = \frac{V_{\mathrm{hull}}}{V_{\max}} = \frac{\sum_{T_i} V_{T_i}}{V_{\max}} \tag{18}$$

where v_{i1}, v_{i2}, and v_{i3} are the three vectors in the tetrahedron T_i, and v_{i1}, v_{i2}, and v_{i3} are edge vectors formed between one vertex of the tetrahedron and each of the other three vertices.

The number of points covered within the agent is computed using Eq. (19) to encourage exploration in regions with higher point density. Additionally, a coverage point increment term is introduced to reduce the reward when a decrease in the number of covered points is observed.

$$R_{\mathrm{points}} = V_{\mathrm{norm}} + \alpha \frac{N_{\mathrm{in}}}{N_{\max}} + \beta \frac{N_{\mathrm{in}} - N_{\mathrm{prev}}}{N_{\max}} \tag{19}$$

where $N_{\max}$ denotes the total number of points in the point clouds, and N_{prev} denotes the number of points covered at the previous step. The parameter α is the weight for the covered-point term and is set to 0.1 in this research, while β is the increment weight and is set to -0.05, with the negative value penalizing a reduction in coverage.

This study further observes that the spherical agent constructed under the reinforcement learning framework is prone to undesirable behaviors such as repeated oscillatory exploration at the same location or becoming trapped in local regions. To address this issue, a region visitation state is incorporated into the reward components to encourage comprehensive exploration. In order to encourage the exploration of areas that have not yet been visited, the point clouds bounding box is uniformly voxelised into a three-dimensional grid with a fixed edge length, denoted by Δ_{grid}. The visitation status of each region is subject to continuous monitoring. This process functions as a guide for the agent's exploration. For each grid region, an exploration reward is assigned according to the visitation frequency, while regions that are visited excessively are penalized. Based on this mechanism, the exploration reward component is formulated as shown in Eq. (20).

$$R_{\mathrm{explore}} = \frac{b}{\sqrt{c_{i'j'k'}}} - \varphi \cdot c_{i'j'k'} \tag{20}$$

where b denotes the exploration incentive coefficient and is set to 0.01 in this research; $c_{i'j'k'}$ represents the historical visitation count of the voxel (i', j', k'); and φ denotes the visitation penalty coefficient, which is set to 0.001.

In order to ensure that the agent can correctly identify the welding joint type specified by the expert knowledge base, the W-LFD descriptors $\Theta_{\min}$, c_l, and r_{ND} are incorporated into a semantic alignment reward. Based on this formulation, the feature matching reward R_{match} is constructed as shown in Eq. (21).

$$R_{\mathrm{match}} = \exp\left(-\left(\lambda_\theta \left(\frac{|\Theta_{\min} - \Theta_{\mathrm{tgt}}|}{\pi} \right)^2 + \lambda_c (c_l - c_{\mathrm{tgt}})^2 + \lambda_r (r_{\mathrm{ND}} - r_{\mathrm{tgt}})^2 \right) \right) \tag{21}$$

where Θ_{tgt}, c_{tgt}, and r_{tgt} denote the target values corresponding to a specific weld joint type. The coefficients λ_θ, λ_c, and λ_r are introduced to balance the sensitivity of different feature dimensions. This reward component reaches its maximum value of 1 only when the observed local features are fully matched with the expert-defined parameters, thereby guiding the agent to converge toward semantically correct weld areas.

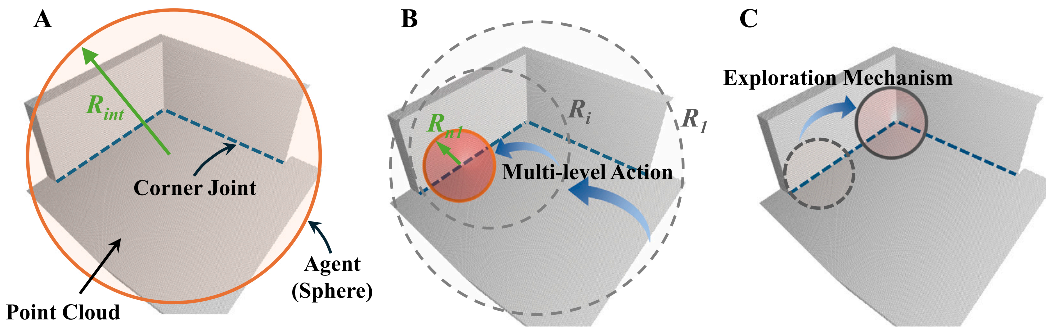


Fig. 6. Schematic diagram of the recognition process based on point clouds exploration mechanism.

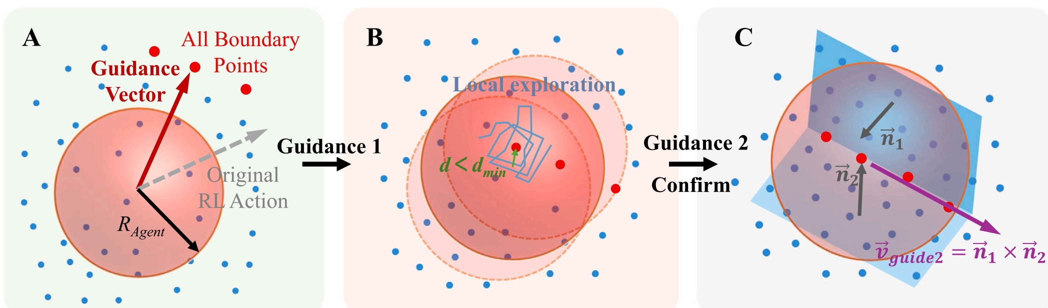


Fig. 7. Schematic illustration of the motion guidance strategy. (A) Searching for edge points within the perception radius. (B) Repeated local exploration after reaching an edge point to verify the target area. (C) Transition from edge-guided navigation to weld-guided navigation.

3.2. Welding area identification

The proposed approach efficiently identifies and localizes welding areas via cooperative optimization by integrating the methods and mechanisms presented in Sections 2 and 3.1. This allows for rapid exploration, comprehensive coverage and precise localization. Furthermore, a motion guidance strategy is proposed to direct the agent in planning the subsequent motion scheme based on its present state. This strategy supports rapid localization during exploration, as well as continuous identification and segmentation of target areas. By embedding physically interpretable feature constraints into the closed-loop intelligent decision-making process, the proposed framework effectively resolves the trade-off between efficiency and accuracy encountered by conventional methods.

After an unmodeled part arrives at the workstation, a robot equipped with a 3D point clouds camera scans the workpiece to acquire raw point clouds data. A background segmentation algorithm is then applied to remove irrelevant information, thereby extracting the part point clouds and constructing the point clouds environment for reinforcement learning, as illustrated in Fig. 5A. Following the provision of the point clouds environment as input, the point clouds exploration mechanism is activated, and the spherical agent dynamically adjusts its position and radius according to the reinforcement learning policy. This process is undertaken to iteratively identify potential welding seam areas in a step-by-step manner. Fig. 5B illustrates the process in which the agent identifies and segments the entire welding area.

Following the procedure shown in Fig. 6, the algorithm employs the multi-level action strategy to store the point clouds of each segmented sub-region in a point clouds repository in real time. When exploration terminates, a union operation is performed over all sub-regions to obtain the complete welding region. Fig. 5C describes how the agent segments the welding region into sub-regions step by step and then combines them into the complete welding region through the union operation. At this stage, the resulting point clouds may exhibit jagged boundaries induced by the spherical geometry of the agent. These boundary irregularities do not affect welding seam localization. Therefore, point clouds filtering is applied as post-processing to fill blank areas and enforce uniformization, and the regularized regional point clouds is used as an accurate input for subsequent welding seam localization algorithms.

To accelerate the exploration of the agent and increase the proportion of effective steps while reducing invalid actions during exploration, a movement guidance strategy is incorporated into the exploration process. This strategy consists of two stages, namely edge-guided navigation and weld-guided navigation. Prior to algorithm execution, all edge points in the preprocessed point clouds are computed in advance. It is important to note that edge point detection is contingent on the presence of abrupt curvature variations. Consequently, the marked points contain edge features exclusively.

After exploration begins, if the agent encounters an edge-marked point within a predefined range, the movement guidance mode is activated, as illustrated in Fig. 7A. In this case, the algorithm determines an optimal geometric guidance vector $\mathbf{v}_{\text{guide}} \in \mathbb{R}^3$ based on the current perceptual state. The system first responds to edge guidance, and the corresponding unit direction vector is defined in Eq. (22). Subsequently, the agent moves according to Fig. 7B and approaches the marked point. It then performs repeated local motions around the marked point to explore the neighborhood multiple times and verify whether the area corresponds to a target area. Once confirmed, the point clouds of this area is segmented. Moreover, since welding areas are typically continuous, we assume that neighboring sub-regions are also continuous along the tangential direction. Based on this assumption, weld guidance is triggered immediately after segmentation, as shown in Fig. 7C. In this step, the vector in Eq. (22) aligns the motion direction of the agent with the cross-product direction between the principal surface normal and the motion direction.

$$\mathbf{v}_{\text{guide}} = \begin{cases} \dfrac{b^* - p_t}{\|b^* - p_t\|} & \exists\, b \in B,\ \|b - p_t\| \le r_t \\ \dfrac{n_1 \times n_2}{\|n_1 \times n_2\|} & \text{Segmented} \wedge t_{\text{seg}} > 0 \end{cases} \tag{22}$$

Algorithm 1 PAEAR: Point Clouds Area Exploration and Active Recognition

Input: Agent state $s = \{p, r\}$, Point clouds P, Boundary points B, Expert Knowledge Φ, Actor Network $\mu(s|\theta^{\mu})$, Thresholds $\delta, r_{\min}$
Output: Union of Segmented Point Clouds $W_{\text{seam}} = \bigcup P_{\text{in}}$

$s_0 \leftarrow \{p_{\text{init}}, r_{\text{init}}\}, \quad F_{\text{guide}} \leftarrow \text{False}, \quad W_{\text{seam}} \leftarrow \emptyset$
while Termination Condition not met ($|B| > 0$) **do**
  $b^* \leftarrow \text{NearestBoundaryDirection}(p, r, B)$
  if $b^* \neq \emptyset$ **or** F_{guide} is True **then**
    if $t_{\text{seg}} > 0$ **then**
      $\mathbf{v}_{\text{guide}} \leftarrow \vec{n}_1 \times \vec{n}_2$
    else
      $\mathbf{v}_{\text{guide}} \leftarrow b^* - p$
    end if
    $a_t \leftarrow T'(\mathbf{v}_{\text{guide}})$
  else
    $a_t^{\text{raw}} = [\eta_t, v_t]^T \leftarrow \mu(s_t)$
  end if
  $a_t^{\text{dec}} \leftarrow \mathcal{I}(\eta_t < \delta) \cdot \mathcal{T}(v_t) + \mathcal{I}(\eta_t \geq \delta) \cdot \mathcal{S}(v_t)$
  $s_{t+1} \leftarrow \text{Execute}(a_t^{\text{dec}})$
  $P_{\text{in}} \leftarrow \{q \in P : \|q - p_{t+1}\| \leq r_{t+1}\}$
  $\mathcal{F}_{\text{geo}} \leftarrow [\Theta_{\min}, c_l, r_{\text{ND}}]$
  $B_{\text{in}} \leftarrow \{b \in B : \|b - p_{t+1}\| \leq r_{t+1}\}$
  if $r_{t+1} \approx r_{min}$ **and** $\mathcal{F}_{\text{geo}} \in \Phi$ **then**
    $W_{\text{seam}} \leftarrow W_{\text{seam}} \cup P_{\text{in}}$
    $P \leftarrow P \setminus P_{\text{in}}$
    $B \leftarrow B \setminus B_{\text{in}}$
    $F_{\text{guide}} \leftarrow \text{True}$
  else
    $B \leftarrow B \setminus B_{\text{in}}; \quad F_{\text{guide}} \leftarrow \text{False}$
  end if
end while
return W_{seam}

where $b^* = \arg\min_{b \in B} \|b - p_t\|$ denotes the nearest boundary point within the current perception range; p_t represents the agent position (x, y, z) at time step t; n_1 and n_2 denote the centroids of the two principal surface-normal clusters, respectively; and r_t denotes the current perception radius of the agent.

In order to ensure compatibility with the action space of the agent, the guidance vector needs to be projected into the spherical action space of the agent. As shown in Eq. (23), we define a kinematic mapping $T' : \mathbb{R}^3 \rightarrow A$ to convert a Cartesian vector into the required azimuth angle ψ and elevation (pitch) angle θ commands. This step ensures that the heuristic guidance signal is compatible with the continuous action space defined for the reinforcement learning policy.

$$\mathcal{T}'(\mathbf{v}) = \begin{bmatrix} \psi \\ \theta \end{bmatrix} = \begin{bmatrix} \text{atan2}(v_y, v_x) \\ \arcsin\left(\frac{v_z}{\|\mathbf{v}\|}\right) \end{bmatrix} \tag{23}$$

During exploration, if the guidance motion mode is enabled, the agent is driven to move according to $\mathcal{T}(\mathbf{v}_{\text{guide}})$. Otherwise, when the guidance motion mode is disabled, the agent continues point clouds exploration following the reinforcement learning policy $\pi_\theta(s_t)$.

Unlike conventional reinforcement learning exploration methods, PAEAR introduces a heuristic guidance strategy based on geometric boundary features and maps it into the agent action space via Eq. (23). 1 summarizes the core procedure of the proposed framework, where efficient active point clouds exploration is achieved through the synergy between physically grounded welding descriptors and the reinforcement learning policy. The corresponding pseudocode is provided below.

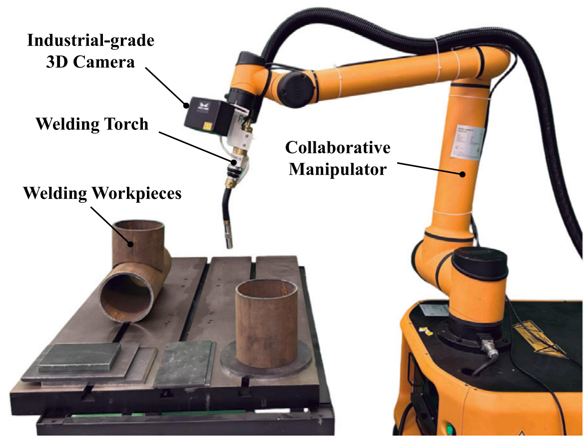


Fig. 8. Robotic welding experimental setup.

4. Experiments

4.1. Experimental preparation

4.1.1. Platform configuration

The proposed algorithm is implemented in Python 3.10. The reinforcement learning component is developed based on the optimization framework of the open-source deep deterministic policy gradient (DDPG) algorithm [37,38]. For other representative reinforcement learning algorithms, methods such as PPO typically require frequent policy updates and large amounts of interaction data to maintain training stability, which increases computational cost in high-dimensional point clouds environments. Alternatively, although methods such as SAC emphasize exploration diversity, the resulting stochastic policies may introduce unnecessary randomness in the later stages of welding seam region exploration. In weld area exploration, the task requires continuous motion control and involves a high-dimensional state space that includes point clouds geometry and bounding box attributes. Under these conditions, the DDPG algorithm can effectively extract complex features. It directly outputs precise continuous actions, demonstrating excellent fitting capability.

All simulation experiments and algorithm deployment are conducted on a 64-bit Windows 11 operating system. The hardware platform is a laptop workstation equipped with an Intel i9-13900HX CPU and an NVIDIA GeForce RTX 4060 GPU. The experimental setup employs an AUBO i16 robotic manipulator and a Mech-Eye NANO 3D camera for physical validation and point clouds acquisition, and the robotic system is assembled as shown in Fig. 8. The same experimental environment is replicated in Gazebo for initial simulation-based data collection. In the simulation environment, the point clouds camera was configured to match the Mech-Eye NANO 3D camera. The resolution was 1280 × 1024, the working distance was 300–600 mm, and the horizontal field of view was 60°. The manipulator provides a payload capacity of 16 kg with a repeatability of ± 0.03 mm. The measurement accuracy of the camera is 0.1 mm at 0.5 m, in accordance with VDI/VDE standards.

In order to validate the effectiveness and generality of the proposed algorithm, four representative samples, namely Sim-1, Sim-2, Real-1, and Real-2, are selected, as shown in Fig. 9. Sim-1 and Sim-2 are constructed from CAD models and evaluated in a ROS-based simulation environment. Real-1 and Real-2 are fabricated as physical workpieces for real-world deployment and verification. These samples differ in complexity: Sim-1 and Real-1 contain only a single welding joint type, while Sim-2 and Real-2 correspond to more complex structures involving multiple welding joint types. The selected parts are different from those in the small-sample dataset introduced later and are mainly

Table 1
Selection of parameters for the weld local feature descriptor.

Weld type	Clustering angle (°)	Curvature sampling radius (mm)	LDCV	NDR
Corner joint	70.0	0.2	0.8	< 0.1
Butt joint	10.0	0.1	0.5	≥ 0.7
Lap joint	85.0	0.1	0.3	≥ 0.1 and < 0.7

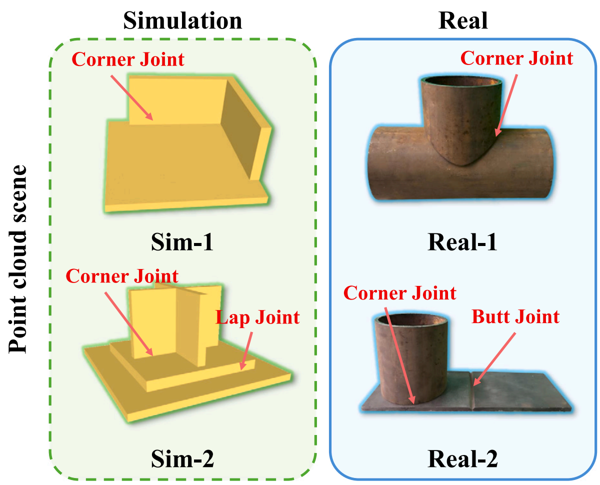


Fig. 9. Workpieces used for experimental validation in simulation and real-world scenarios.

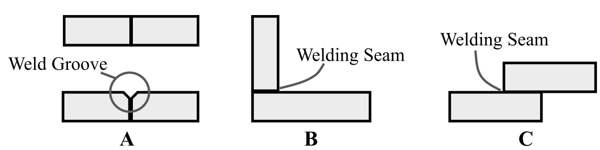


Fig. 10. Three weld joint types used in the experiments. (A) Butt joint (B) Corner joint (C) Lap joint.

composed of typical welding seam combinations from the primary dataset. This design aims to examine the recognition performance and adaptability of the proposed method as well as the baseline methods.

4.1.2. Parameter selection

As shown in Fig. 10, to evaluate the performance of PAEAR across different weld joint types and under varying joint conditions within the same welding process, three common welding configurations are considered: butt joints, corner joints, and lap joints.

The weld local feature descriptor (W-LFD) established in Section 2 characterizes different weld joint types through variations in its parameters. Precise tuning of three descriptor parameters is required to identify a single weld joint or multiple weld joints. Table 1 lists the specific configurations corresponding to different joint types in detail. In multi-type recognition scenarios, the PAEAR framework operates in a segmented manner according to the weld joint type. Accordingly, dynamic weights are applied to the parameters at different stages; for example, during corner weld recognition, the reward function is positively correlated with LDCV and negatively correlated with NDR.

4.1.3. Dataset construction

PAEAR requires a small-sample dataset covering multiple weld joint types for pre-training. Accordingly, this paper presents 10 datasets of commonly used welding parts frequently adopted for evaluation in related research. The datasets are categorized by joint type and include 5 corner-joint samples, 3 butt-joint samples and 2 lap-joint samples. Notably, the corner-joint samples include point clouds data collected from complex scenarios, such as intersecting joints and curved trajectories. To enhance the model's generalization capability to real-world environments, we also collect data from 4 actual industrial components. Although the dataset is small in scale, it provides strong category representativeness. Using it as the testing benchmark not only covers common welding joint types but also highlights the performance of the proposed algorithm under low-data-resource conditions. Table 2 presents schematic illustrations, annotated point clouds, and detailed specifications of the samples. In addition, this dataset serves as the training baseline for the comparative methods.

4.1.4. Model pre-training

In order to ensure reproducibility of the experiments, an actor-critic network architecture for point clouds exploration learning is implemented based on the PyTorch framework. Both the actor network and the critic network adopt a multilayer perceptron (MLP) structure. In terms of architecture, the actor network consists of an input layer, two hidden layers, and an output layer. The input layer receives the high-dimensional state vector s_t. Each hidden layer contains 64 hidden units and uses the ReLU activation function for nonlinear mapping. The output layer uses the Tanh activation function to constrain the action outputs to the range $[-1, 1]$. An action bound tensor is then applied to map these outputs to the physical action space. Specifically, this corresponds to an azimuth angle in $[-\pi, \pi]$ and a pitch angle in $[-\pi/2, \pi/2]$. The critic network follows a similar design. The input layer is responsible for concatenating the state s_t and the action a_t. The hidden layers also utilize 64 units with ReLU activations, and the output layer is a fully connected layer without an activation function, directly outputting the predicted Q-value.

Regarding the training configuration, the detailed hyperparameter settings are listed in Table 4.

The effectiveness of the exploration policy during the pre-training stage is evaluated by recording the evolution of key reinforcement learning variables throughout model pre-training. Pre-training is conducted using representative parts with typical welding joint types from the dataset. During this stage, the repeated confirmation procedure is not applied after reaching a target point. Instead, the detected area is directly trusted. It is then identified as the target area. In this study, the metric variables obtained during pre-training on a representative corner-weld sample are selected for further analysis.

As demonstrated in Fig. 11A, during the training phase, the agent employs the PAEAR strategy, concurrently reducing the search radius. This results in a significant decrease in the number of covered points prior to segmentation. This phenomenon is reflected in Fig. 11B by negative values of coverage efficiency, confirming that the proposed algorithm tends to localize the target efficiently rather than performing indiscriminate traversal. After segmentation is completed, resetting the radius causes a surge in the number of covered points, resulting in periodic peaks in the efficiency curve. In addition, the decrease in navigational efficiency indicates the cost of non-straight trajectories incurred by the agent when traversing complex areas. As can be observed from Fig. 11C, value function learning remains generally stable: the mean TD error fluctuates around 0 with bounded variance, and occasional spikes quickly subside. In the initial phase, severe oscillations of approximately ± 0.9 reflect estimation bias. As training progresses, the amplitude of the oscillations gradually decays toward 0. This indicates

Table 2
Description of the small-sample dataset.

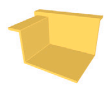
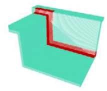
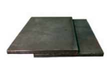
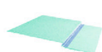
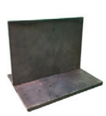
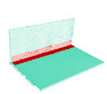
Model View	Labeled Point Clouds	Data Source	Point Clouds Size / Labeled Points	Description
		Simulation	75305 / 9223	Corner joint
		Simulation	69066 / 4803	Corner joint with intersecting line trajectory
		Simulation	34019 / 5413	Complex corner joint
		Simulation	27161 / 3179	Butt joint
		Simulation	47523 / 5638	Butt joint
		Simulation	47687 / 1571	Lap joint
		Real	683710 / 66428	Butt joint
		Real	483766 / 35209	Lap joint
		Real	344108 / 48041	Corner joint
		Real	209762 / 25301	Corner joint with curved welding path

Table 3
Comparison of multiple methods.

Algorithm	mIoU (%)	mER ± SD	mCR (%)	Total samples	Discriminative power
PAEAR(Ours)	**86.86**	**0.054 ±0.047**	**100.0**	**4**	**High**
DGCNN	5.64	0.840 ± 0.147	62.5	10	Low
PointNet	11.56	0.789 ± 0.244	50.0	10	Low
PointNet2-MSG	61.91	0.149 ± 0.109	100.0	10	Medium
PointNet2-SSG	32.21	0.140 ± 0.258	87.5	10	Medium

that the value network is converging over time. Moreover, the TD-error histogram exhibits a peak at 0, which further verifies the absence of significant systematic bias in the network.

Fig. 11D-F present the statistical characteristics of the agent actions along the three spatial axes. The solid lines represent the mean values, showing the directional preference of the actions. The dashed lines represent the standard deviations, which quantify the exploration intensity. The results show that, in the early stage of training, the standard deviations along all three axes remain at relatively high levels, indicating that the policy is in a high-intensity exploration phase. In the convergence stage, the standard deviations of the y- and z-axis components gradually decrease to near 0 and become stable, while the mean of the x-axis component approaches 0. This suggests that the policy has converged to an efficient trajectory. Overall, the learned policy tends to advance smoothly with small steps, with its randomness and oscillations effectively controlled.

Figs. 12 and 13 depict the loss convergence and reward evolution during training, respectively. As shown in Fig. 12, the critic loss exhibits

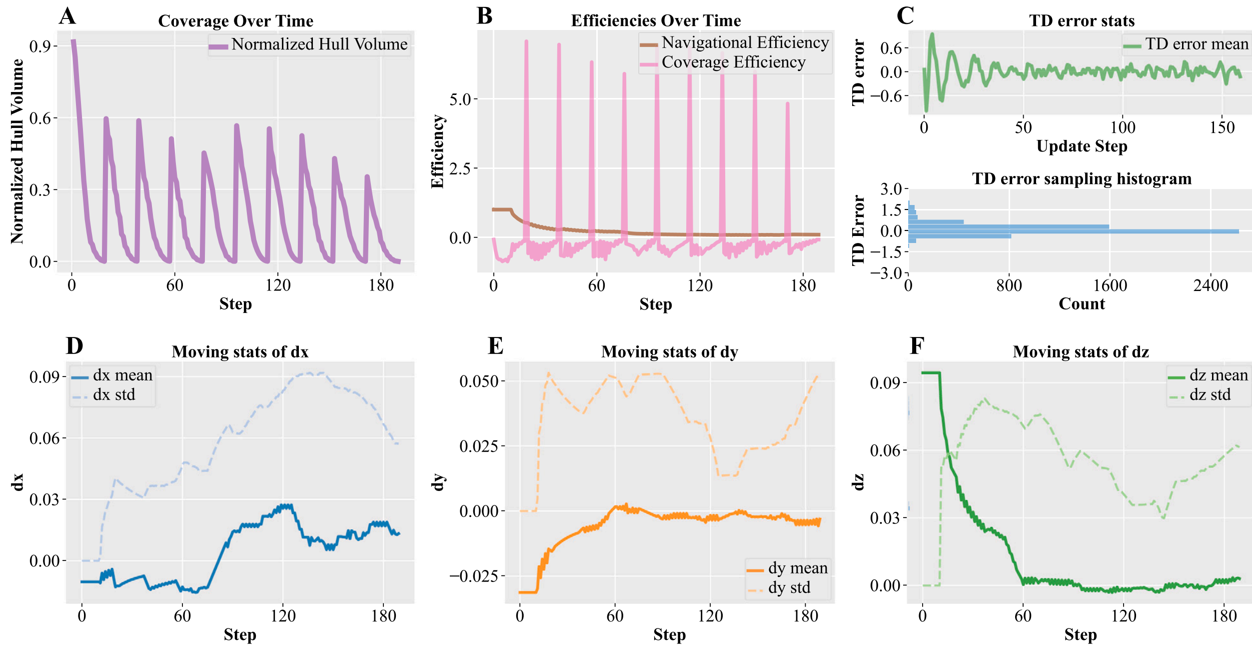


Fig. 11. Curves related to the pre-training process.

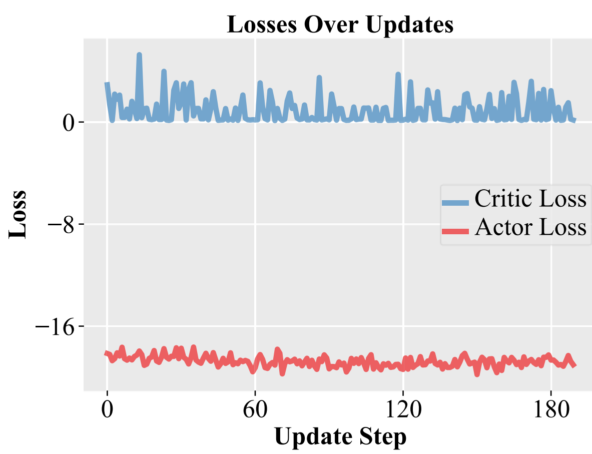


Fig. 12. Training loss.

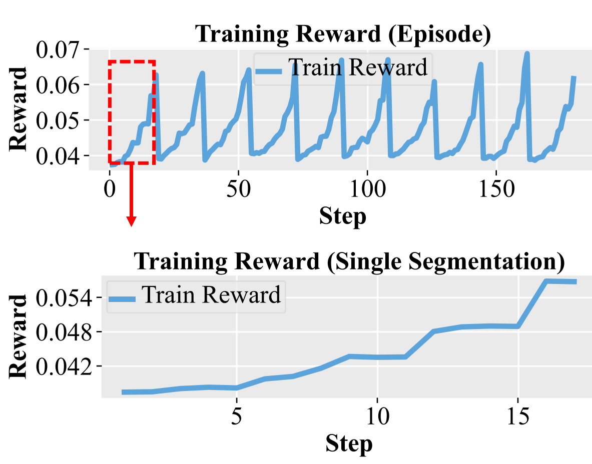


Fig. 13. Training reward.

Table 4
Hyperparameters of the reinforcement learning method in this research.

Parameter	Value	Description
Optimizer	Adam	Optimization algorithm
lr_{actor}	3×10^{-4}	Learning rate of actor network
lr_{critic}	3×10^{-3}	Learning rate of critic network
N_{hidden}	64	Number of neurons in hidden layers
γ	0.98	Discount factor
τ	0.005	Soft update coefficient
N_{batch}	100	Batch size
C_{buffer}	1000	Capacity of replay buffer
σ_{exp}	0.01	Standard deviation of Gaussian noise

bounded oscillations centered around 0, while the actor loss stably converges to approximately −18, indicating robust policy updates without divergence. As demonstrated in Fig. 13, the dynamic characteristics of reward acquisition are further illustrated. The periodic sawtooth pattern in the upper plot corresponds to the training mechanism that performs multiple segmentations within a single episode, with peaks remaining at a consistently high level. The lower plot shows that, within a single segmentation task, the reward increases monotonically in a stepwise manner. These observations confirm that the agent can effectively accumulate returns through consecutive correct decisions, thereby achieving rapid convergence.

4.2. Comparative experiment

In order to validate the overall performance of PAEAR under small-sample conditions, this section selects several classical deep learning networks in point clouds processing as reference methods for comparative experiments, including PointNet [39], DGCNN [40], PointNet++MSG [41], and PointNet++SSG [41]. Quantitative evaluation is conducted using four metrics: Intersection over Union (IoU), Error Rate (ER), Category Recall (CR), and Total Samples. The specific definitions

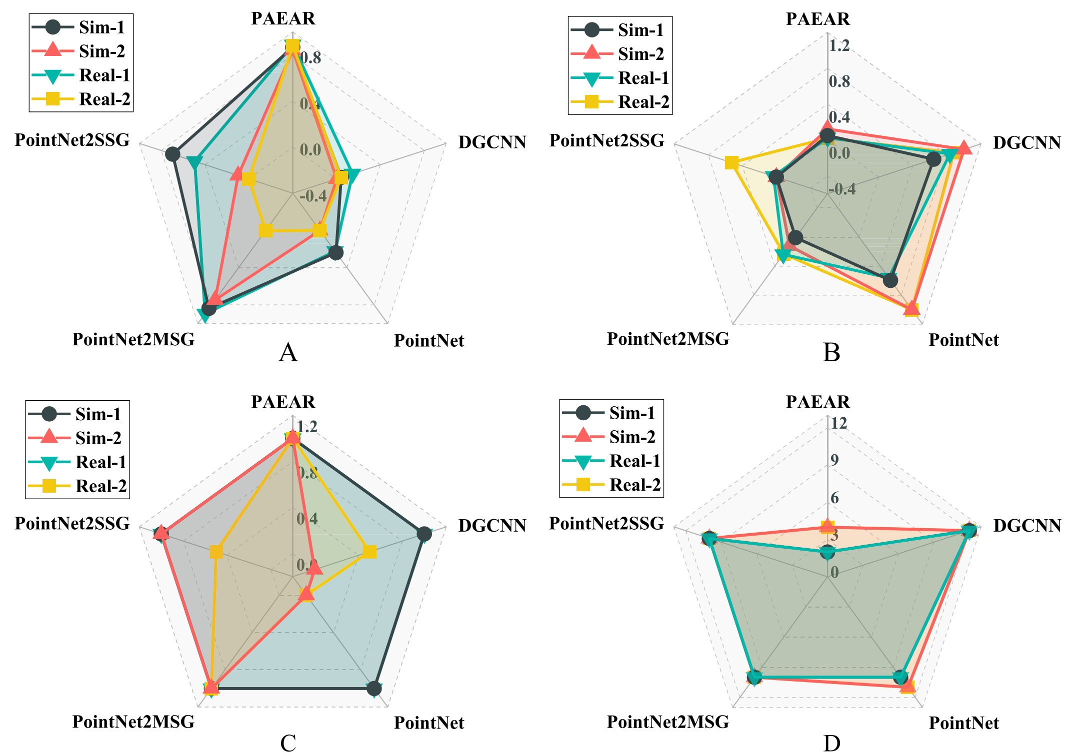


Fig. 14. Performance comparison of different methods on four point clouds scenarios. (A) intersection over union (IoU). (B) Error rate (ER). (C) Category recall (CR). (D) Total Samples.

are given in Eqs. (24)–(26). Moreover, the results over the four scenarios are averaged to obtain mIoU, mER, and mCR, as reported in Table 3 and Fig. 14.

$$\mathrm{IoU}_{\mathrm{sc}} = \frac{P_T^{\mathrm{sc}}}{P_{Ld}^{\mathrm{sc}} + P_F^{\mathrm{sc}}} \times 100\% = \frac{P_T^{\mathrm{sc}} \cdot P^{\mathrm{sc}}}{P_d^{\mathrm{sc}} \cdot P_L^{\mathrm{sc}} + P_F^{\mathrm{sc}}} \times 100\% \tag{24}$$

$$\mathrm{ER}_{\mathrm{sc}} = \frac{P_F^{\mathrm{sc}}}{P_{Rec}^{\mathrm{sc}}} = \frac{P_F^{\mathrm{sc}}}{P_T^{\mathrm{sc}} + P_F^{\mathrm{sc}}} \tag{25}$$

$$\mathrm{CR}_{\mathrm{sc}} = \frac{C_T^{\mathrm{sc}}}{C^{\mathrm{sc}}} \times 100\% \tag{26}$$

where sc denotes the scenario index used to distinguish the four predefined experimental scenarios (Sim-1, Sim-2, Real-1, and Real-2). P_T^{sc} and P_F^{sc} denote the numbers of correctly and incorrectly identified points in scenario sc, respectively. It should be noted that, during recognition, all methods are executed in a downsampled point clouds environment generated from the raw point clouds. P_{Ld}^{sc} denotes the number of labeled reference points after downsampling in scenario sc. It is computed by mapping the labeled reference points in the raw point clouds, denoted as P_L^{sc}, into the downsampled point clouds. The computation depends on the raw point count P^{sc} and the downsampled point count P_d^{sc}. $P_{\mathrm{Rec}}^{\mathrm{sc}}$ denotes all points recognized by the algorithm in scenario sc. C_T^{sc} denotes the number of correctly recognized categories, and C^{sc} denotes the total number of welding seam categories contained in scenario sc. Finally, mIoU, mER, and mCR are obtained by summing the corresponding metrics over all scenarios and dividing by the number of scenarios.

As indicated by the statistics in Table 3, the proposed PAEAR method demonstrates a clear advantage even when trained with only 4 samples. Across the four scenarios, PAEAR achieves an mIoU of 86.86% and an mER of 0.054, while the CR reaches 100% in all scenarios, substantially outperforming the other methods. Notably, PointNet++MSG can attain an IoU comparable to PAEAR in certain scenarios, and its CR can also reach 100% across the four scenarios; however, it requires more samples and yields a higher ER. These results suggest that the reinforcement learning-based active perception strategy can effectively drive the agent to traverse the complete welding seam area. In contrast, conventional deep learning methods, which lack explicit modeling of local weld-seam geometric characteristics, often struggle to extract continuous and complete welding seam morphology when confronted with complex unannotated background noise.

This paper presents a detailed quantitative comparison of four metrics for five methods across four scenarios. To provide a more intuitive visualization of the overall performance of each method, a comprehensive radar chart is plotted as shown in Fig. 14. As shown in Fig. 14A and C, PAEAR achieves higher intersection over union (IoU) and category recall (CR) than the other methods across all four scenarios. As shown in Fig. 14B, the proposed method also yields a lower error rate (ER) than the other methods in all four scenarios.

As shown in Fig. 14D, other comparison algorithms typically require more than 10 training samples to converge and are prone to overfitting, whereas PAEAR leverages the pre-trained expert knowledge descriptor (W-LFD), which substantially reduces its dependence on data volume and improves sample efficiency. As can be observed, PAEAR exhibits higher robustness across all four scenarios than the other methods. By comparison, PointNet++MSG performs slightly worse than PAEAR, while PointNet++SSG shows large performance variations across different scenarios, indicating limited robustness and adaptability. In addition, DGCNN and PointNet tend to produce a large number of outlier misclassifications, leading to a significantly increased error rate.

In addition, the proposed method is compared with various approaches from the perspectives of geometric methods, sensor-based recognition, and programming strategies. In terms of trajectory accuracy, it is compared with the method of Rout et al. [42], where the reference trajectory is established based on manual-assisted coarse screening and high-precision cross-sectional fitting and smoothing [47], and accuracy is evaluated as the maximum absolute error, defined by the Euclidean distance from each recognized point to its nearest reference point. Furthermore, the proposed method is compared with the approaches of Rout et al. in 2020 [43] and 2021 [44] in terms of the

Table 5
Comparison results with other algorithms under different parameters.

Baseline method	Evaluation metric	Baseline result	PAEAR (Ours)
Rout et al. [42]	Trajectory accuracy	0.28 mm	0.19 mm
Rout et al. [43]	Mean end-effector deviation	3.18%	1.54%
Rout et al. [44]	Mean end-effector deviation	2.04%	1.54%
Yang et al. [45]	Confusion rate	58.93%	6.67%
Deepak et al. [46]	CAD-Free Capability	No	Yes

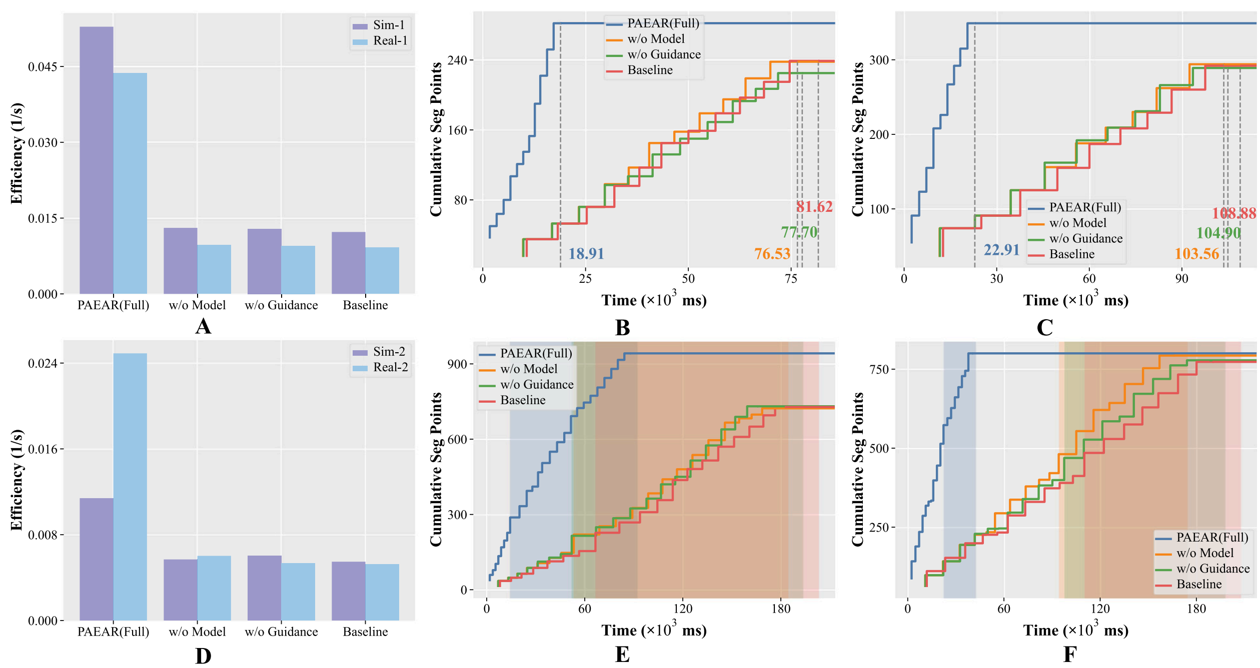


Fig. 15. Ablation Comparison Experiments for PAEAR. (A) Comparison of efficiency in a single weld joint scenarios. (B) Time-series plot of segmented point count in the Sim-1 scenario. (C) Time-series plot of segmented point count in the Real-1 scenario. (D) Comparison of efficiency across multiple weld joint scenarios. (E) Time-series plot of segmented point count in the Sim-2 scenario. (F) Time-series plot of segmented point count in the Real-2 scenario.

average deviation of the end-effector position. The theoretical position is defined as the coordinates sent to the robot via ROS, while the actual position is obtained from robot feedback. The average deviation D_{mean} is computed as shown in Eq. (27). In this equation, $\mathbf{P}_i$ denotes the position vector of the ith reference point on the theoretical trajectory, expressed as $(x_{t,i}, y_{t,i}, z_{t,i})$, and $\mathbf{Q}_i$ denotes the position vector of the ith evaluation point on the actual trajectory after arc-length interpolation-based resampling, strictly aligned with $\mathbf{P}_i$, expressed as $(x_{a,i}, y_{a,i}, z_{a,i})$. In terms of confusion rate, the proposed method is compared with the algorithm of Yang et al. [45], where the confusion rate is defined as the ratio of non-weld edges or boundary lines incorrectly classified as weld trajectories to the total number of detected edges. Additionally, a qualitative comparison is conducted with the method of Deepak et al. [46] regarding dependency on 3D models.

$$D_{\text{mean}} = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{P}_i - \mathbf{Q}_i\|_2 \tag{27}$$

Based on the comparative results in Table 5, it can be concluded that the proposed PAEAR algorithm outperforms existing baseline methods across all key evaluation metrics. Without relying on predefined CAD models, the proposed method achieves a trajectory accuracy of 0.19 mm and maintains the average end-effector position deviation at 1.54%, effectively reducing motion planning and execution errors. In addition, during the feature extraction stage, the proposed method achieves a confusion rate of only 6.67%. These results indicate that the proposed approach not only eliminates the dependency on 3D models but also ensures high-precision trajectory generation and robust feature recognition, demonstrating significant advantages and strong potential for practical applications in complex industrial environments.

4.3. Ablation experiment

In order to assess the individual contributions of the movement guidance and the pre-trained model components to the overall PAEAR framework, we conduct an ablation study in four simulated and real-world environments (Sim-1, Sim-2, Real-1, and Real-2). The evaluation compares the complete PAEAR algorithm (Full) with three variants: a version without movement guidance (w/o Guidance), a version without the pre-trained model (w/o Model), and a baseline exploration framework that excludes both components (Baseline).

Fig. 15AD show that the complete PAEAR algorithm can accomplish the exploration, identification, and segmentation of all target welding seam point clouds areas with the highest efficiency in both single weld joint scenarios and multiple weld joint scenarios. The efficiency metric is defined as the reciprocal of the total time required by each method to identify and segment all target point clouds. Across the scenarios Sim-1, Sim-2, Real-1, and Real-2, the efficiency of PAEAR is improved by an average of 75.95%, 49.76%, 78.34%, and 77.87% compared with w/o Guidance, w/o Model, Baseline, and Baseline, respectively.

Fig. 15 plot the number of segmented points over time for the four methods in the four scenarios (Sim-1, Real-1, Sim-2, and Real-2), together with the completion time required to finish recognition. As can be seen from Fig. 15BC, in single weld joint scenarios, the complete PAEAR algorithm identifies a larger number of target points within a shorter time. In Sim-1 and Real-1, the final number of segmented points obtained by PAEAR is increased by an average of 20.51% and 19.66% compared with the other methods, respectively. After removing action guidance and the pre-trained model, the recognition efficiency

Table 6
Ablation study of the proposed algorithm.

Algorithm	Scene	Single segmentation time (ms)	Path efficiency	Number of identified points	Step
PAEAR (Full)	**Sim-1**	**1576.17**	**0.366**	**1693**	**12**
	Sim-2	**3371.87**	**0.294**	**8403**	**26**
	Real-1	**2290.87**	**0.246**	**2081**	**10**
	Real-2	**2113.28**	**0.330**	**4590**	**19**
w/o Guidance	Sim-1	6475.35	0.197	1456	12
	Sim-2	8278.19	0.199	4382	20
	Real-1	10 490.27	0.088	1676	10
	Real-2	9908.61	0.145	4554	19
w/o Model	Sim-1	6377.91	0.198	1525	12
	Sim-2	8359.78	0.115	5010	21
	Real-1	10 356.30	0.090	1666	10
	Real-2	9214.95	0.085	4390	18
Baseline	Sim-1	6801.63	0.198	1523	12
	Sim-2	8684.01	0.192	4649	21
	Real-1	10 887.85	0.090	1658	10
	Real-2	10 557.78	0.128	4292	18

decreases noticeably. The dashed lines and the associated numbers indicate the completion time of each method. As demonstrated in Fig. 15EF, the employment of PAEAR in weld joint scenarios consistently yields advantageous outcomes. In Sim-2 and Real-2, the number of segmented points increases by 19.51% and 2.39% relative to alternative methods, respectively. In addition, the shaded background regions with the same color as the corresponding curves indicate the identification and segmentation process for the second welding joint type.

In this study, the performance of PAEAR is systematically compared with multiple ablation variants and baseline methods. The comparison is conducted in both simulation (Sim) and real-world (Real) scenarios using four key metrics: single-step segmentation time, path efficiency, the number of recognized points, and the number of segmentation steps. The results in Table 6 indicate that, benefiting from the complete model architecture and the guidance strategy, PAEAR (Full) achieves the best overall performance across all test scenarios. In particular, the single segmentation time of PAEAR (Full) is only about 20%–30% of that of the other variants, demonstrating markedly improved computational efficiency. Meanwhile, in terms of path efficiency, the full method also significantly outperforms the variants with missing components. Conversely, the performance of w/o Guidance and w/o Model exhibits a substantial decline, approaching that of the Baseline. This suggests that the removal of both components leads to an increased propensity for unguided and purposeless exploration in the point clouds environment.

4.4. Recognition effect and analysis

In order to provide a more intuitive analysis of the performance of different methods in practical welding scenarios, Fig. 16 visualizes the point clouds recognition results under four representative scenarios. As can be observed, in the simulation scenario Sim-1, DGCNN exhibits under-segmentation, while PointNet identifies only part of the target area but overall suffers from over-segmentation. Both the proposed PAEAR and the two PointNet++ variants can recognize the relevant areas, although PointNet++SSG shows slight under-segmentation. In the Sim-2 scenario, neither DGCNN nor PointNet can identify the areas corresponding to the two welding joint types. PointNet++SSG is able to detect both areas but displays severe under-segmentation, whereas PointNet++MSG and PAEAR can both achieve complete segmentation of the target areas.

In scenario Real-1, DGCNN demonstrates severe over-segmentation, while PointNet and PointNet++SSG exhibit under-segmentation. Conversely, both PointNet++MSG and the proposed PAEAR can accurately identify and segment the relevant point clouds areas. In the more complex real-world scenario Real-2, PointNet fails to recognize the target areas, and DGCNN shows severe over-segmentation. PointNet++SSG identifies only one welding joint type and presents pronounced under-segmentation. PointNet++MSG can recognize and segment both welding joint types, but still exhibits under-segmentation. By comparison, PAEAR achieves complete identification and segmentation of both welding seam point clouds areas in this scenario.

High-quality point clouds recognition is the foundation for automated welding, as it directly enables reliable welding trajectory planning. Based on the above recognition results, we further validate the robot welding trajectory generation performance in the real-world scenarios Real-1 and Real-2. To verify the usability of the point clouds segmented by the proposed method for physical welding, a relatively simple fitting and boundary-point computation approach is adopted to localize the welding seam position within the local point clouds.

As shown in Figs. 17 and 18, in practical scenarios, the robot acquires the workpiece point clouds from a random viewing angle and directly transmits the point clouds data to the simulation environment via ROS topics. The PAEAR algorithm then subscribes to the corresponding point clouds topic to read the data, performs downsampling in the simulation environment, and completes weld area identification and segmentation. Based on the segmented results, a straightforward boundary-point computation and fitting method is adopted to generate a smooth welding trajectory. In addition, a statistical analysis is conducted on the time distribution and computational efficiency across three stages: point clouds acquisition, active perception, and motion planning. For the real-world scenarios Real-1 and Real-2, the computational efficiency is illustrated by the curves shown in Fig. 19. The results demonstrate that the overall cycle time meets the real-time requirements of practical manufacturing scenarios. This cycle includes point clouds acquisition in the real environment, simulation-based planning, and execution by a real robot. Moreover, the time variation across different scenarios is relatively small, indicating that the proposed method can adapt to complex environments while maintaining high computational efficiency in recognition and segmentation.

It should be noted that PAEAR essentially serves as a scalable front-end perception module that can be integrated with various welding seam localization algorithms. Its core function is to segment and classify local welding seam point clouds areas. In the above experimental validation, welding seam localization is performed using a relatively straightforward boundary-point computation approach. This approach has limited stability under complex working conditions. In practical applications, more mature and robust localization algorithms tailored to different welding joint types can be incorporated on top of the local welding seam point clouds segmented by PAEAR. This results in further improvements to the accuracy and stability of welding seam localization. In addition, reinforcement learning outcomes may exhibit uncertainty. The associated risks can be mitigated by imposing more stringent segmentation verification criteria to constrain accuracy. For

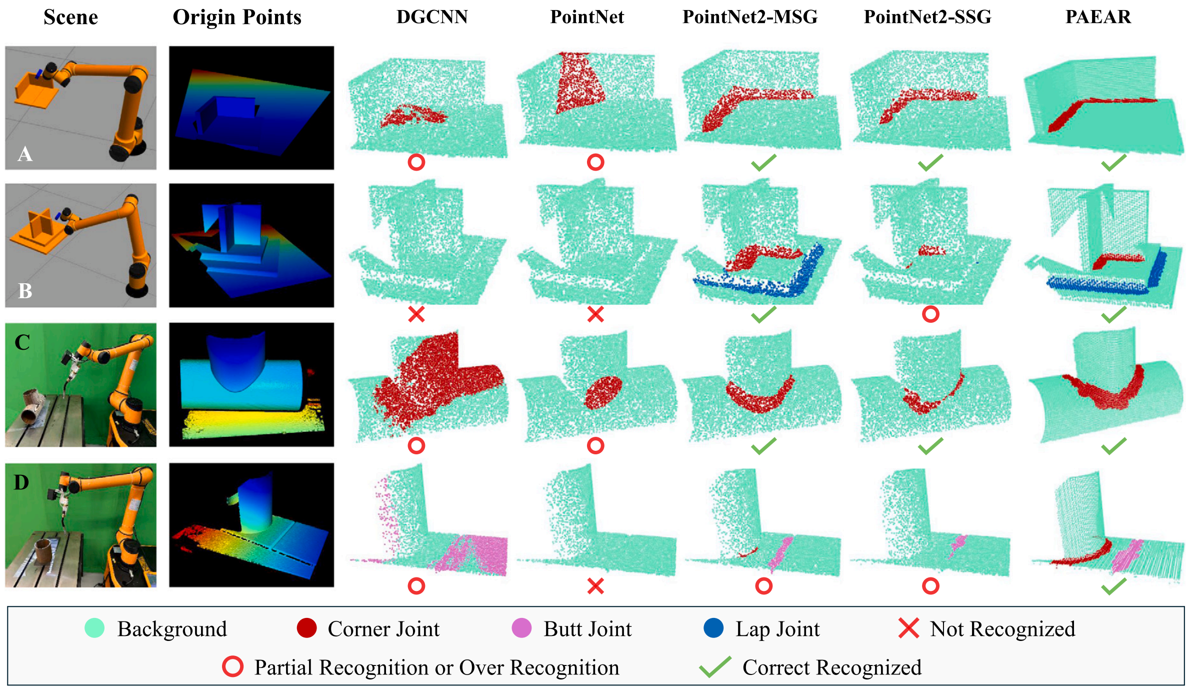


Fig. 16. Point clouds recognition results of different methods under four scenarios.

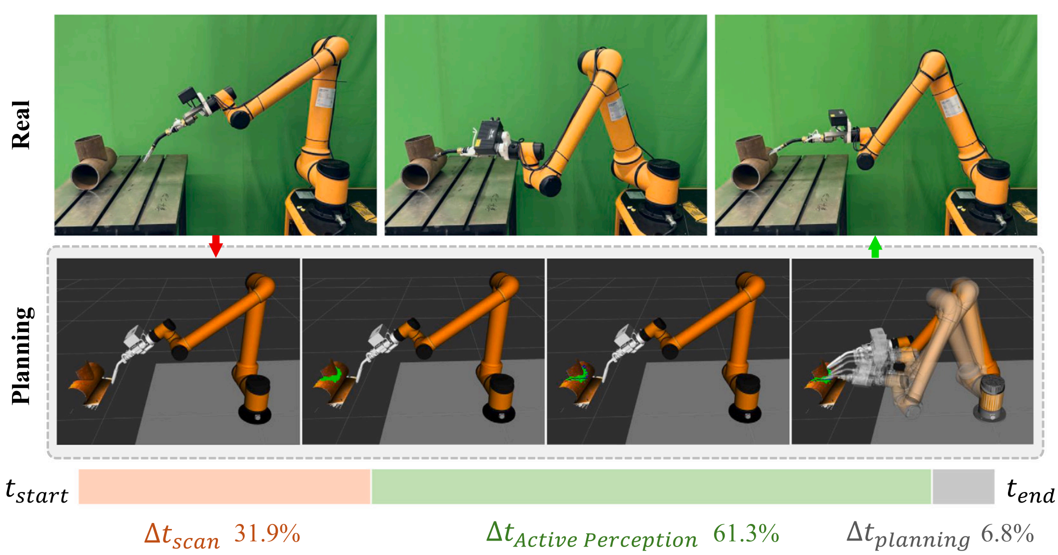


Fig. 17. Welding trajectory planning results of PAEAR in the Real-1 scenario.

motion planning in robotic welding, safety risks at the planning level can be addressed in ROS by importing part point clouds captured by the camera from the real environment and by integrating components such as MoveIt!. This simulation-based planning process inherently accounts for critical robot kinematic constraints, including inverse kinematics, joint limits, and singularity avoidance, ensuring that the generated paths are safely executable. The planned trajectory is finally sent to the real robot for welding on the physical workpiece.

5. Discussion

After the spherical agent constructed in this study segment and reorganize the point clouds, wavy boundaries may appear along the segmentation edges. This phenomenon does not affect subsequent welding seam localization, because accurate seam localization fundamentally depends on the intermediate topological features of the point clouds rather than the geometric regularity of the outer boundaries of the segmented patches [48,49]. As long as PAEAR ensures that the Agent-enveloped region fully covers the actual welding seam, these wavy boundaries can be ignored during centerline extraction. To further ensure welding seam localization performance, filtering and point clouds reconstruction methods will be introduced in future work to eliminate irregular edge features.

As a perception module, PAEAR can be integrated with more advanced welding seam localization algorithms to achieve more precise recognition. It can also be combined with localization algorithms for different joint types, thereby enabling the localization of multiple types of welding seam. Such extensibility is of considerable significance for industrial applications. Although the current PAEAR framework can accurately identify welding regions under various geometric conditions, it mainly serves as a front-end perception module and is not yet capable of extracting precise joint depth values to guide downstream welding

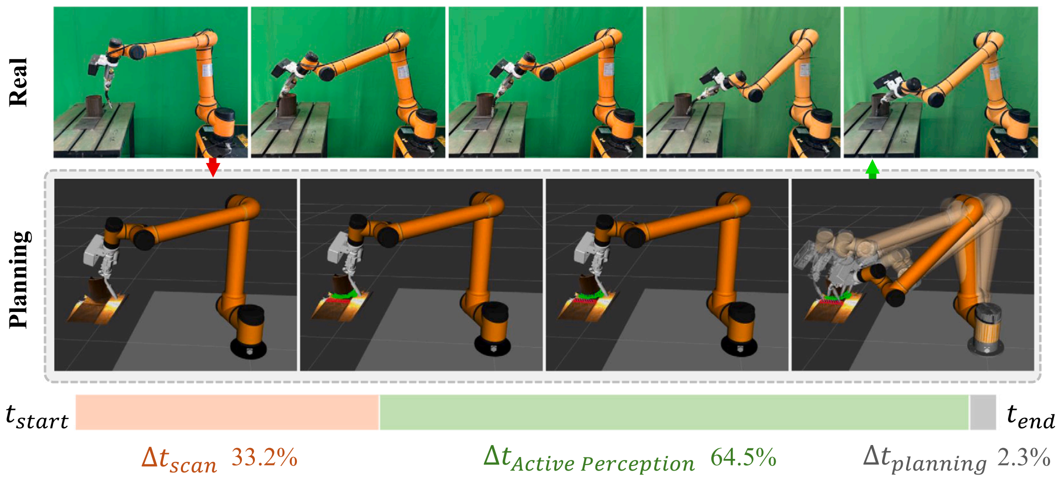


Fig. 18. Welding trajectory planning results of PAEAR in the Real-2 scenario.

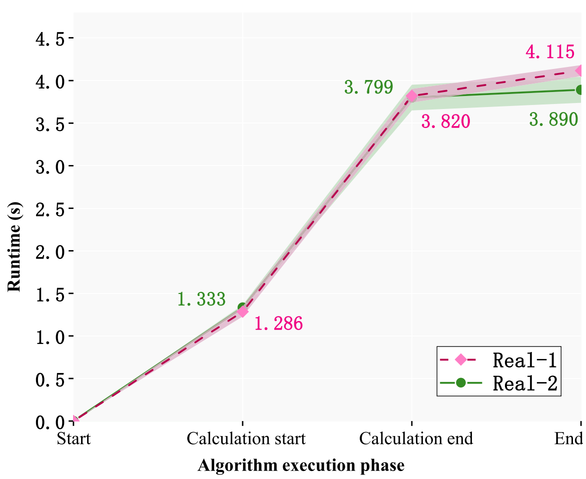


Fig. 19. Computational efficiency curves in real-world scenarios.

process parameters, such as electrode deposition rate. Integrating depth measurement algorithms into the segmented point clouds to achieve dynamic process control will therefore be a major focus of future research.

Although PAEAR is validated on a laptop workstation, it is portable to industrial embedded platforms. The network adopts a lightweight multilayer perceptron architecture, and deploying a pre-trained model avoids the computational burden of on-device training. Although active exploration requires continuous point clouds processing, modern embedded hardware platforms generally provide sufficient computational capability for such tasks. In addition, ROS natively supports distributed deployment. PAEAR can operate as an independent perception node on a robot controller running ROS and seamlessly transmit localization results to the controller.

6. Conclusion

In order to enable robots to interpret the complex geometry of unmodeled workpieces, this paper proposes an active exploration paradigm that combines local point clouds descriptors with reinforcement learning. Small-sample pre-training is used to extract expert priors, which are parameterized and embedded into the reinforcement learning state space. A multi-level action strategy guides the agent through the full process from coarse search to fine-grained perception. This enables autonomous welding of unmodeled workpieces in unstructured environments. PAEAR serves as an intelligent perception framework for welding robots and enables multi-type welding seam area recognition on unmodeled workpieces. It overcomes the limitations of conventional edge-based approaches that lack weld joint category information and operate with low efficiency. It also supports classification and discrimination among seam, edge, and surface elements. The proposed algorithm provides strong scalability and generalization capability. It can serve as a front-end perception module that integrates with various localization methods. This enables efficient classification and localization of complex welding seam for different unfamiliar parts within a single process.

In the process of welding seam recognition, PAEAR can identify and classify welding seam areas in new and unfamiliar workpiece scenarios that differ from the training samples. Across the four new scenarios, an mIoU of 86.86% is achieved, representing an improvement of 24.95% to 81.22% over the deep learning baselines. The average recognition error rate is as low as 0.054, with a maximum reduction of 0.786 compared with the other methods. Across the four scenarios examined, we achieved 100% category recall for all welding joint types. In single weld joint scenarios, incorporating guidance and the pre-model increases efficiency by an average of 77.14% and improves the number of recognized points by 20.09%. In multiple weld joint scenarios, efficiency improves by 63.81% on average, accompanied by a 15.95% increase in the number of recognized points.

CRediT authorship contribution statement

Yong Tao: Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Donghua Tan:** Writing – original draft, Visualization, Validation, Methodology, Investigation. **Fan Ren:** Writing – review & editing, Supervision, Investigation, Formal analysis. **Pai Zheng:** Writing – review & editing, Supervision, Conceptualization. **Jiewu Leng:** Writing – review & editing. **Yazui Liu:** Writing – review & editing, Validation, Formal analysis, Conceptualization. **Wei Wang:** Writing – review & editing, Supervision, Conceptualization. **Hongxing Wei:** Writing – review & editing, Supervision, Conceptualization. **Lihui Wang:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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